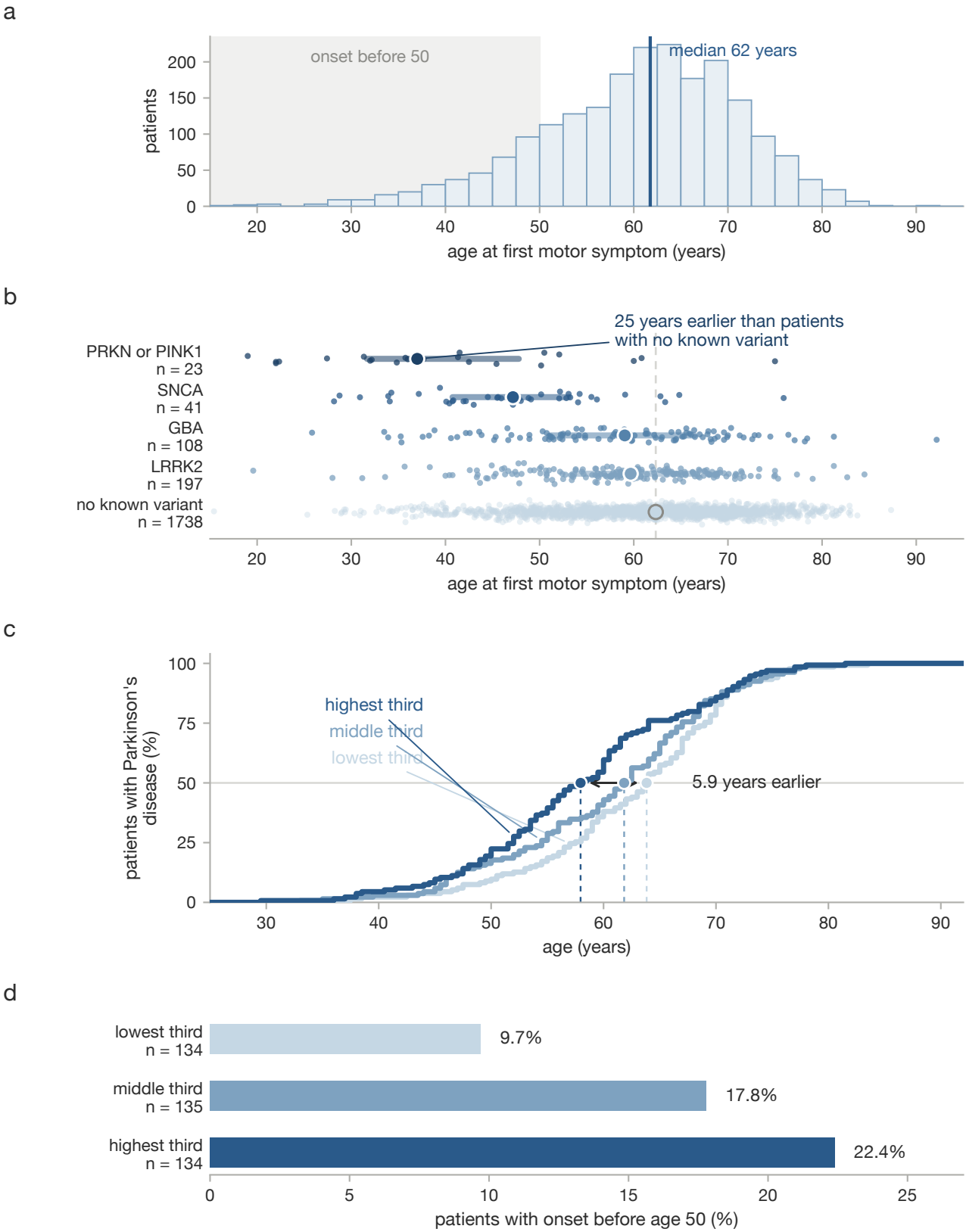
a
onset before 50
median 62 years
patients
200
150
100
50
0
20
30
40
50
60
70
80
90
age at first motor symptom (years)
b
25 years earlier than patients with no known variant
PRKN or PINK1 n = 23
SNCA n = 41
GBA n = 108
LRRK2 n = 197
no known variant n = 1738
20
30
40
50
60
70
80
90
age at first motor symptom (years)
c
patients with Parkinson's disease (%)
100
75
50
25
0
highest third
middle third
lowest third
5.9 years earlier
30
40
50
60
70
80
90
age (years)
d
lowest third n = 134
9.7%
middle third n = 135
17.8%
highest third n = 134
22.4%
0
5
10
15
20
25
patients with onset before age 50 (%)